Day-ahead Forecasting for Redispatch Measures using Machine Learning Models

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Outline

1. Motivation
2. Forecaster setup
3. Data
4. Methodology
5. Empirical results
6. Robustness check
7. Interpretability
8. Conclusions



Transmission Network

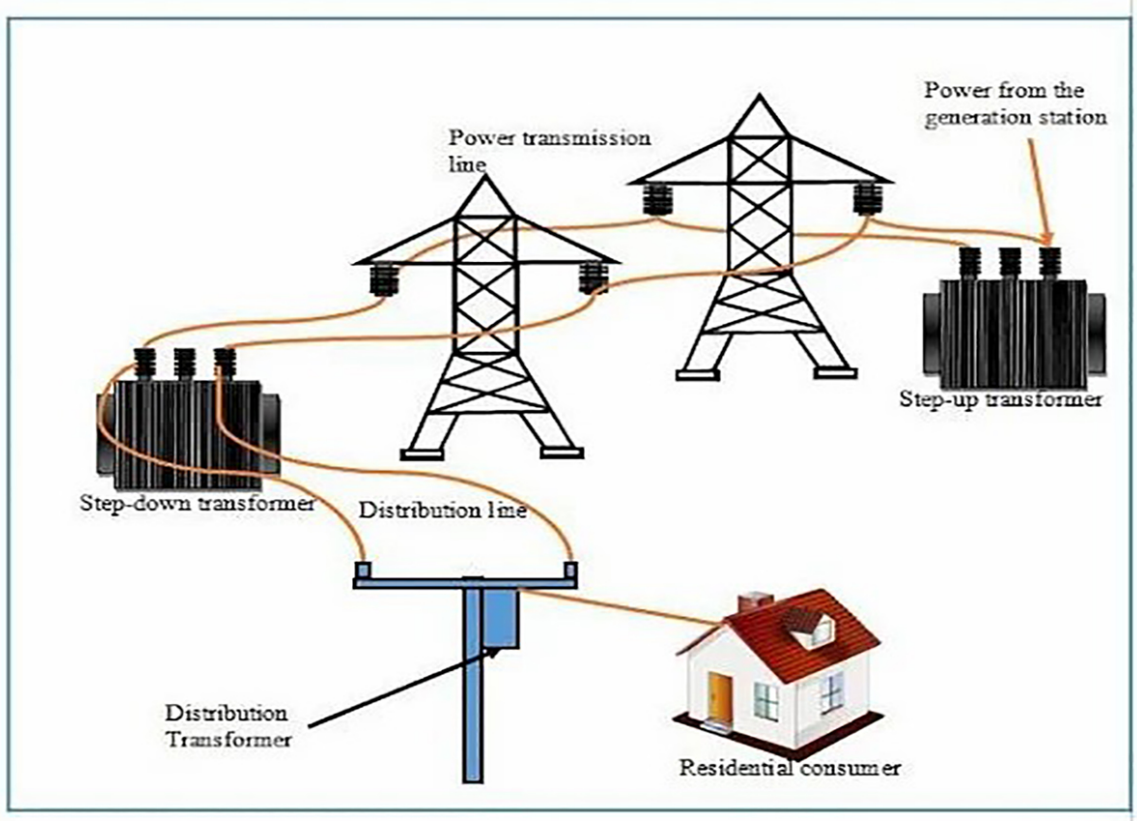


Figure 1: An overview of electricity transmission. From generation facilities to final consumers, **Transmission Control Operators** (TSOs) are in charge of the process. Source: https://www.whatoop.com/power-transmission-system-101-guide/



German TSOs

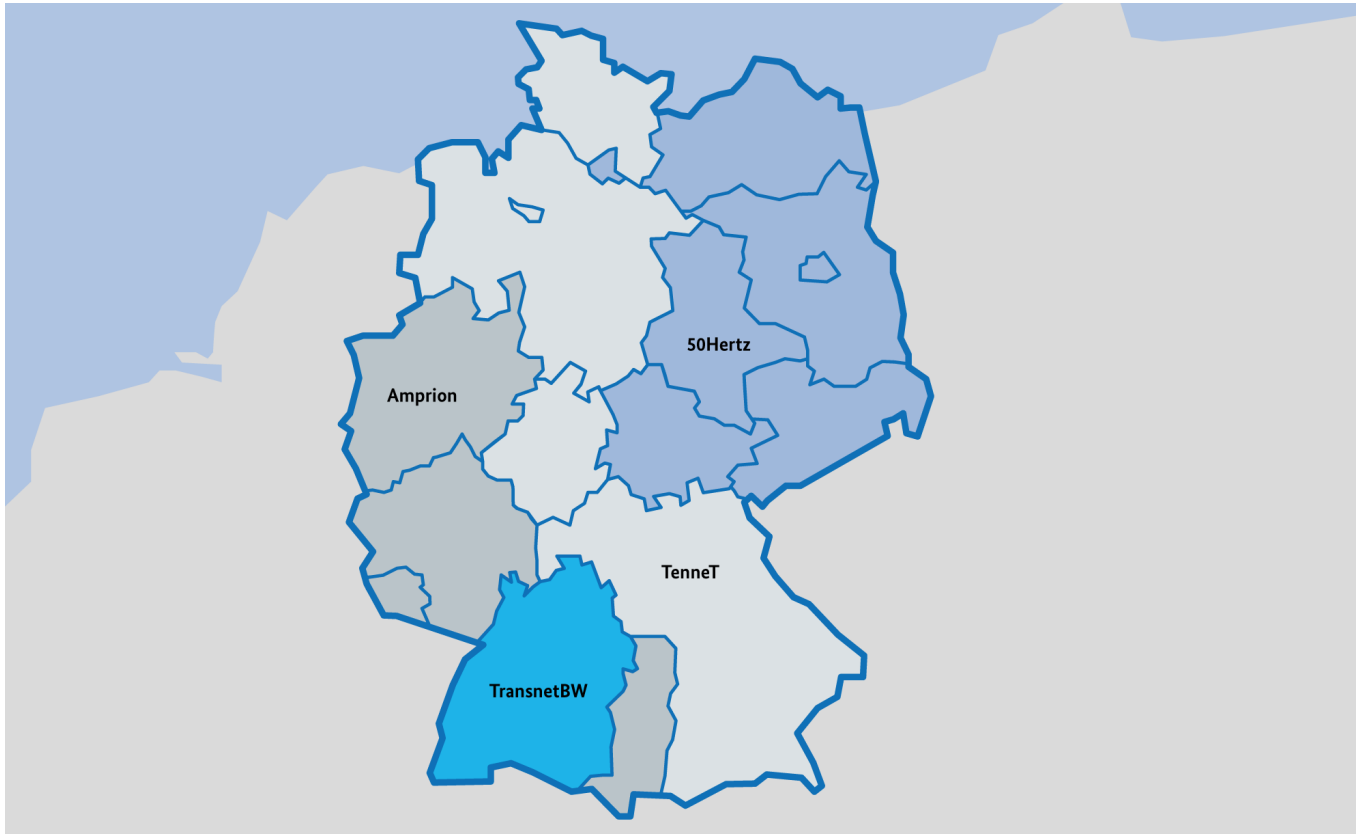


Figure 2: German TSOs with geographical administrative areas: TenneT DE, 50Hertz, Amprion, and TransnetBW. Source: (Bundesnetzagentur, n.d.)



Congestion management

- Prevents line overloads
- Locational (activated resources must reside within a confined area)
- A few strategies are possible, depending on the national market design and regulation
- Because of high market connectivity, TSOs from multiple countries must coordinate when needed to avoid blackouts - see the Czech snapped line as an example of chain reactions that cause outage (Reuters, 2025)



Germany major bottleneck

- North: onshore and offshore wind farms $\rightarrow$ high supply

$\Downarrow$

- South: heavily industrialized $\rightarrow$ high demand

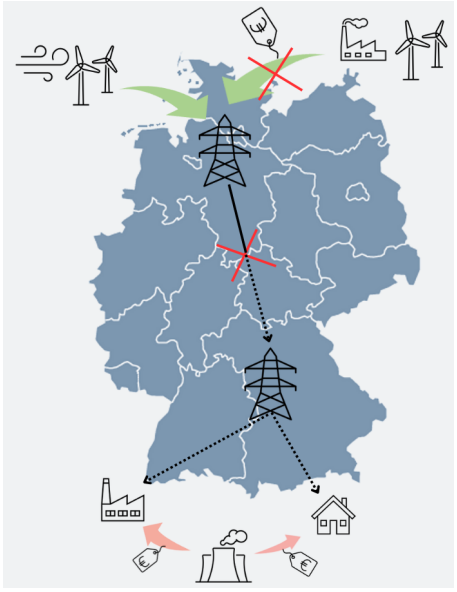


Figure 3: North-South bottleneck in Germany's transmission system. RES sources in the North must travel all the way South before the largest part is absorbed



Redispatch in Germany

- A **cost-based redispatch** scheme
- After the day-ahead market, TSOs begin a network planning phase
- Using projected data, grid sections with high probability congestion (bottlenecks) are identified
- Certain installations are instructed to increase or decrease their output to line integrity $\rightarrow$ **up and down interventions**
- Reimbursement is offered after-the-fact



Redispatch depiction

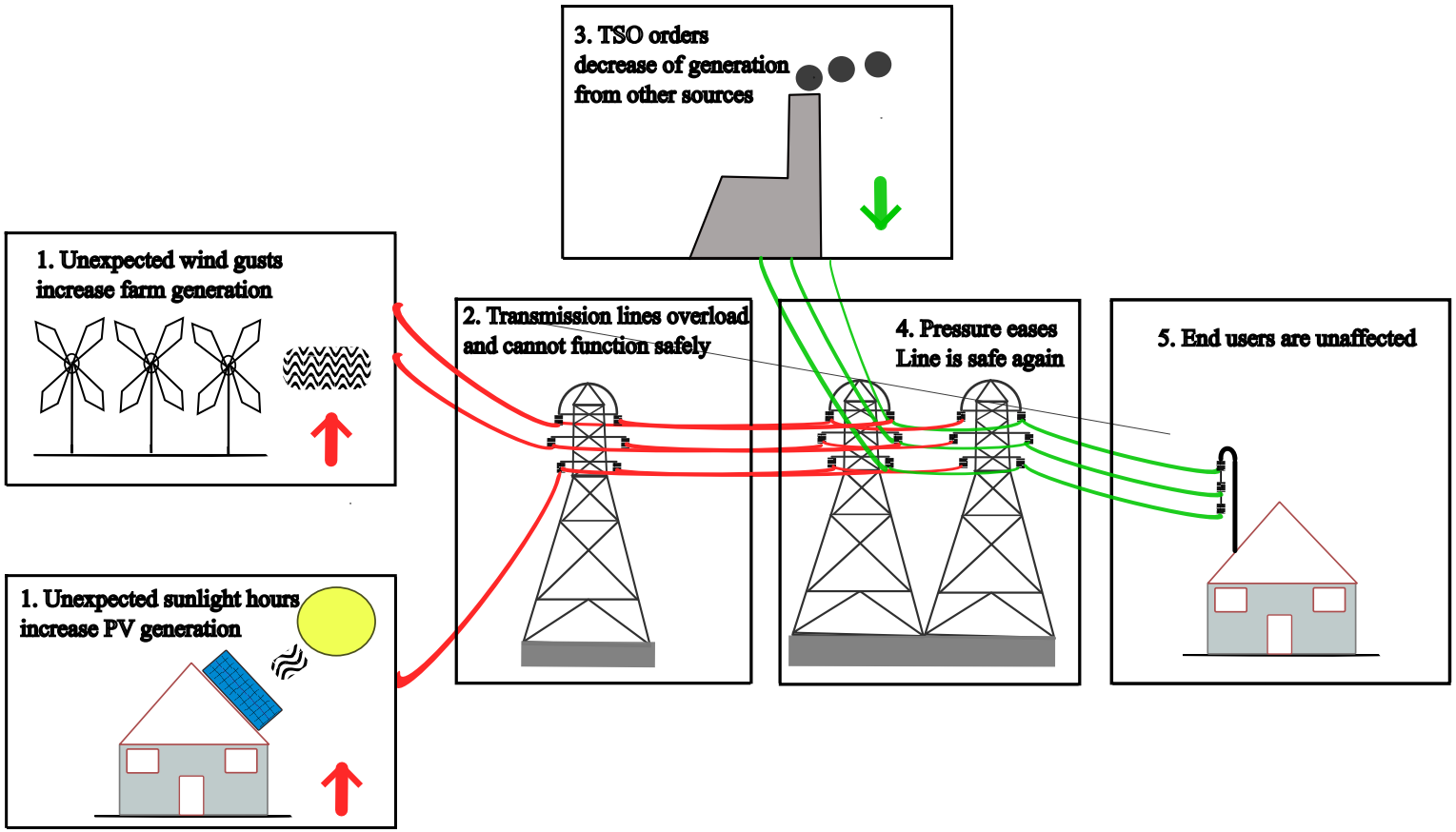


Figure 4: The redispatch approach to congestion in Germany. Downward case.



Redispatch estimation

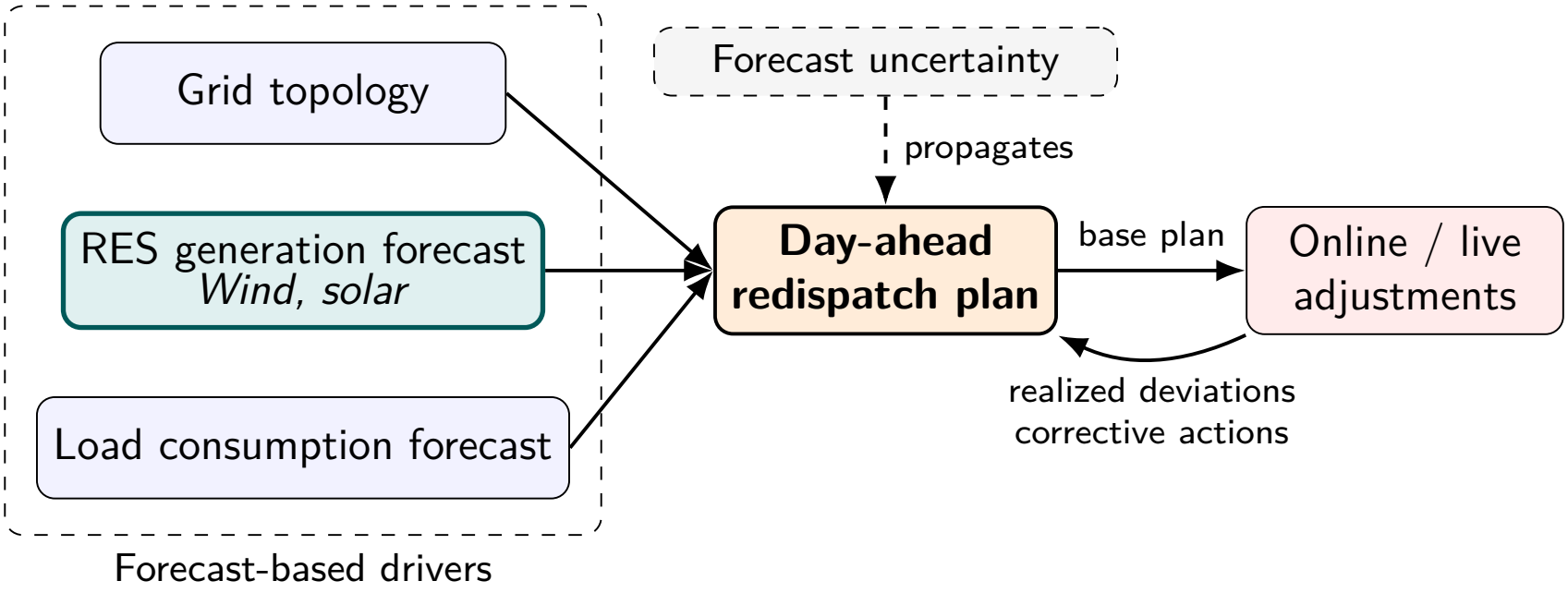


Figure 5: Redispatch as a forecast-based planning problem under uncertainty.



Research questions

1. *Do neural multi-horizon forecasters beat strong statistical/ML benchmarks in direct redispatch forecasting?*

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2. *Do exogenous covariates significantly improve performance?*

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3. *Is their predictive power driven by correct temporal alignment?*



Related work

- Event prediction (probability of redispatch): (Alali et al., 2021; Staudt, Rausch, et al., 2019; Staudt, Traeris, et al., 2018)
- Operational optimization / decision support: (Kloubert et al., 2015; Liao et al., 2025)
- Ex-post analysis and explainability: (Titz, Pütz, & Witthaut, 2024)



What we do compared to others?

- Comparing with existing studies:
  - *Advantages*: Complementary to operational grid simulations and potentially faster at inference time
  - *Disadvantages*: Possibly not as accurate, GPU acceleration necessary for efficient training of neural models
- Some studies develop an economic cost analysis, which requires node-level data
- We do not compare with current practices because (i) they are not public and (ii) public data is too broad



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Prediction problem

- Multi-horizon time series regression with exogenous inputs
- Denote y_t as hourly redispatch volume (MW), then:

$$\hat{y}_{t+1:t+H} = f\left(y_{t-L+1:t},\ x^{hist}_{t-L+1:t},\ x^{futr}_{t+1:t+H}\right)$$

- We predict a horizon of H future time steps using L historical observations (input size)
- x^{hist} are historical covariates (available up to time t), while x^{futr} are known at prediction time (day-ahead forecasts, schedules)



Further details

- One model is trained per TSO, jointly predicting up/down directions: 4 model instances per architecture.
- Day-ahead forecasting ($H = 24$), consistent with operational planning
- We predict only with information available at D-1 afternoon

▸ Information Set



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Target construction

- Source: (Netztransparenz, 2025)
- Final target: **hourly** redispatch volume (MW), per TSO and direction
- Construction:
  - Filter within-Germany intervention types
  - Hourly split by overlap weighting ▸ Details
  - Assign to directing TSO and sum by direction
  - Fill missing hours with 0
- Period: 01.10.2021 - 20.02.2026



Descriptive statistics

TSO	Direction	Mean	Median	Q_{90}	Q_{99}	Sparsity
50Hertz	down	339.8	5.9	1139.0	3128.2	49.44%
	up	25.1	0.0	24.3	486.8	87.51%
Amprion	down	56.2	0.0	179.3	1013.4	84.16%
	up	476.7	0.0	1596.5	3551.7	50.31%
TenneT DE	down	732.3	255.8	2195.8	4915.7	31.56%
	up	154.8	6.4	462.6	1243.5	47.38%
TransnetBW	down	10.3	0.0	10.0	287.1	86.19%
	up	367.9	93.0	1162.5	2395.9	42.14%

Table 1: Descriptive statistics of the target variable by TSO and direction. Full dataset.



Target characteristics

- Strong sparsity: many zero periods, particularly in some datasets
- Right-skewed and heavy-tailed: large redispatch events dominate the mean, while typical values remain low
- Temporal clustering: redispatch occurs in sustained episodes (average episode duration lasts 6–7 hours hours in our dataset)
- Non-linear and regime-dependent behavior



Features

- **System state (forecasts and actuals)**
  - Wind, solar, generation, consumption
- **Market signals (forecasts)**
  - Electricity price spreads
- **Grid interactions**
  - Scheduled cross-border exchanges (forecasts)
- **Runlength & data-driven**
  - Constructed from data associated with the (lagged) target
  - Rolling-based episode aggregates
  - Power weighted by active duration ratio
- For actuals, 24h lags and 7 day rolling means were used
- All features and target are standard-scaled
- Source: ENTSO-E Transparency



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Model selection

- **Criteria**:
  - Well-known benchmarks
  - Tabular model for capturing non-linearities without a clear temporal structure and exogenous feature/covariate support
  - Neural forecasters with first-class covariate integration
- **Choices**:
  - Persistence/statistical: Naive (last value), ARIMA
  - Linear/tabular: Ridge regression, Seasonal Ridge, LightGBM
  - Neural sequence/decomposition: LSTM, NBEATSx, NHiTS, TFT

▸ NBEATSx ▸ NHITS ▸ TFT



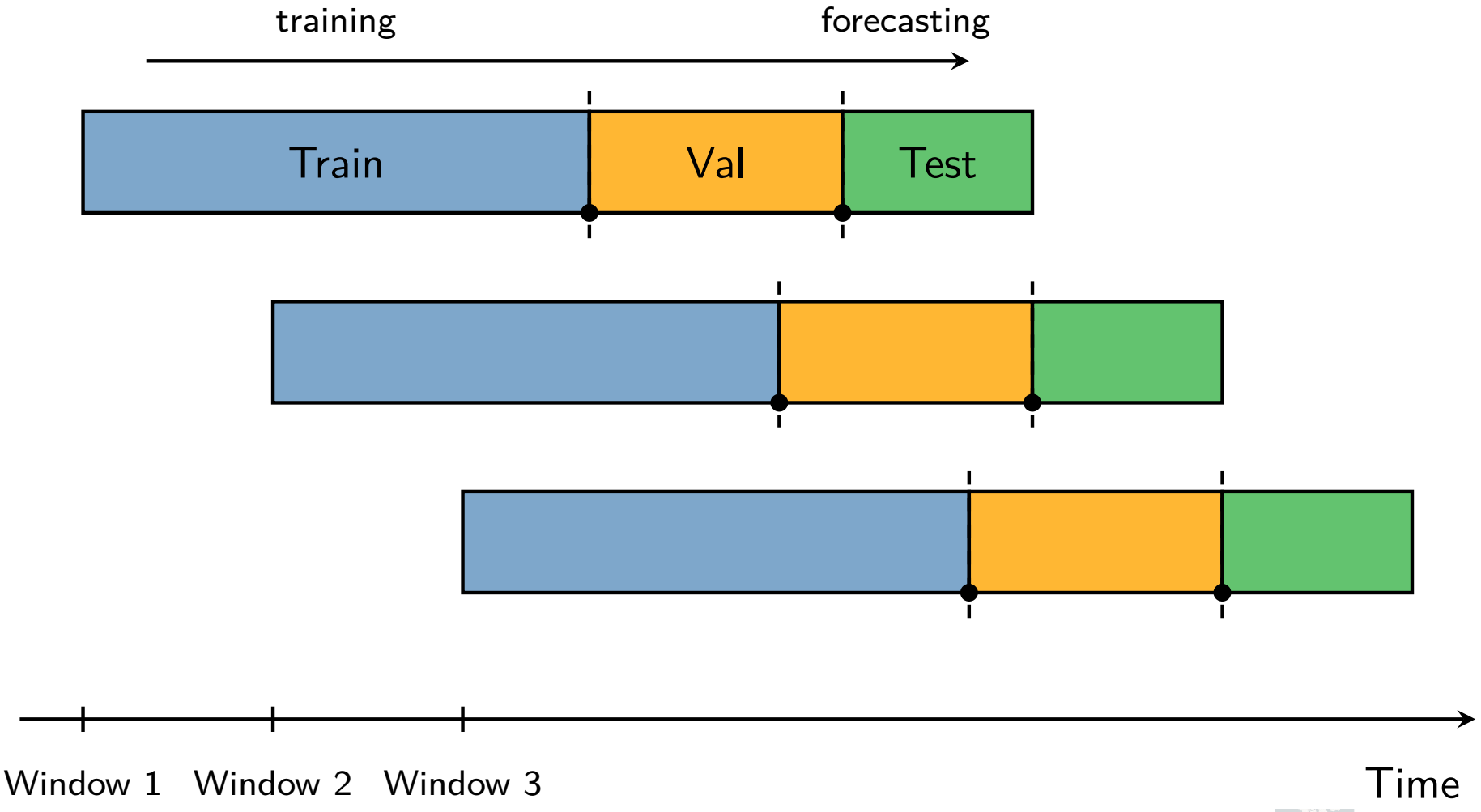
Training overview

- All models are trained in a gated rolling window approach, where each window includes a train split of 37 months, a validation of 2 months (used for early stopping), and a testing period of 1 month.
- For example, the first window models are trained on the period [2021-10-01, 2025-01-01), with early stopping on [2025-01-01, 2025-03-01), and prediction for [2025-03-01, 2025-04-01).
- The total test period amounts to [2025-03-01, 2026-02-20]. In total, there are 11 test windows

▸ Hyperparameter tuning



Gated rolling window evaluation



Evaluation metrics

- Magnitude is evaluated with *Mean Absolute Error* (MAE):
  - More robust to outliers, which add difficulty to the prediction task with marginal gains
  - Supported for other energy time series tasks (Olivares et al., 2021)
- Occurrence detection is assessed using the $F_{\beta=2}$ score for the event $\mathbf{1}(y > 0)$:
  - We prioritize recall over precision because, particularly for upwards redispatch, missing events can lead to more drastic and costlier interventions
- MAE-based evaluation constitutes the primary model selection criterion, and $F_{\beta=2}$ as a complementary diagnostic



Multiple model comparison

- Model Confidence Set (MCS) for MAE (Hansen, Lunde, & Nason, 2011)
- Block bootstrap-based paired superiority test for $F_{\beta=2}$
- Diebold-Mariano (DM) tests for robustness, where only 2 alternatives are compared at a time (Diebold & Mariano, 1995)



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Evaluation metrics

Dataset details			Models				
TSO	Metric	Dir.	LightGBM	LSTM	NBEATSx	NHiTS	TFT
50Hertz	MAE	down	202.4	199.3	**165.1**	172.4	173.4
		up	30.2	31.6	33.1	**28.6**	28.9
	$F_{\beta=2}$	down	0.82	0.83	0.83	0.83	**0.84**
		up	0.20	0.36	0.51	0.48	**0.52**
TenneT DE	MAE	down	506.7	393.6	**324.6**	336.7	339.3
		up	93.3	97.6	90.3	88.4	**84.1**
	$F_{\beta=2}$	down	0.89	0.89	0.87	0.87	**0.89**
		up	0.80	0.80	0.78	0.78	**0.82**

Table 2: Out-of-sample evaluations per model, whole period for 50hertz and TenneT DE. Best value(s) per row are bolded.



Evaluation metrics

Dataset details			Models				
TSO	Metric	Dir.	LightGBM	LSTM	NBEATSx	NHiTS	TFT
Amprion	MAE	down	80.8	81.5	**70.9**	71.4	76.5
		up	424.6	381.7	302.6	**298.9**	302.0
	$F_{\beta=2}$	down	0.09	0.45	0.54	0.55	**0.59**
		up	0.76	**0.82**	0.79	0.80	0.81
TransnetBW	MAE	down	18.0	18.5	**15.1**	15.7	16.4
		up	230.4	219.1	**165.2**	168.3	176.6
	$F_{\beta=2}$	down	0.37	0.60	0.63	**0.64**	0.54
		up	**0.87**	0.84	0.87	0.87	0.87

Table 3: Out-of-sample evaluations per model, whole period for Amprion and TransnetBW. Best value(s) per row are bolded.



Predictions zoom-in

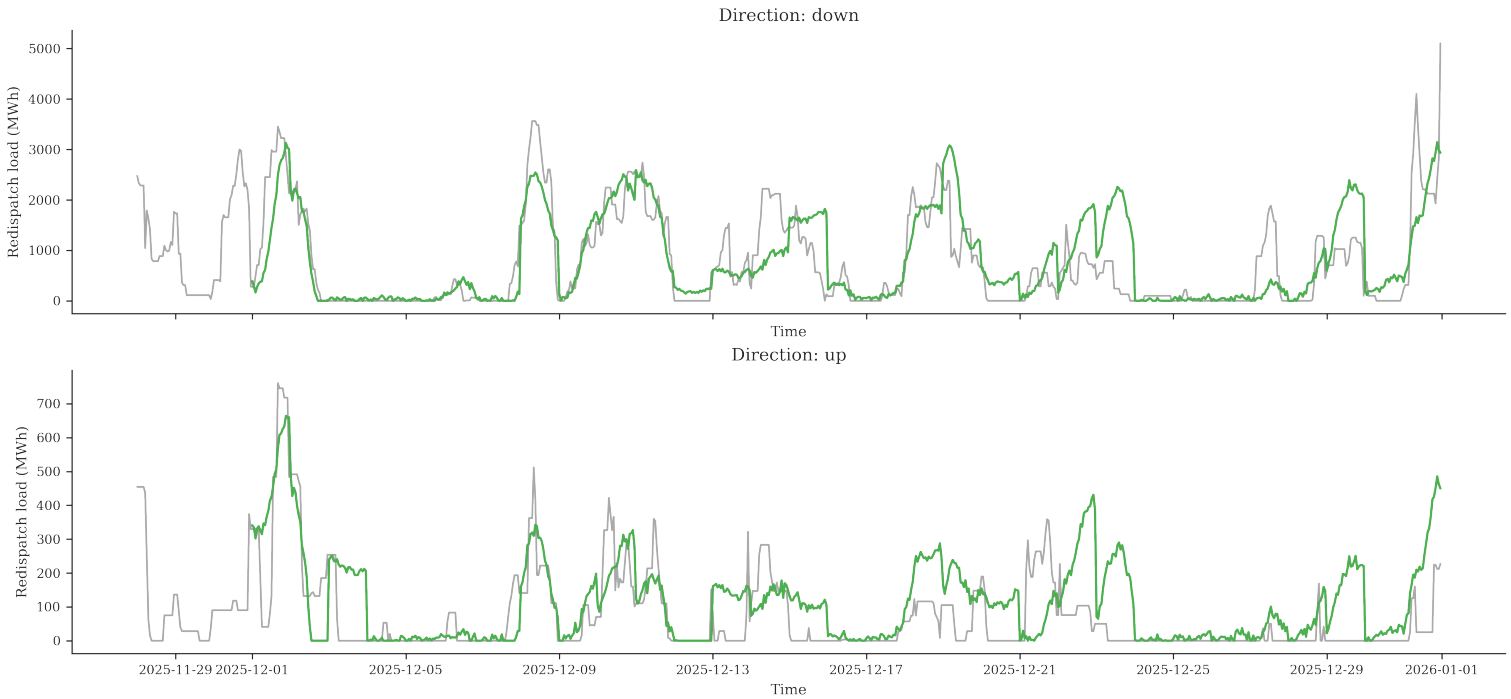


Figure 6: NBEATSx predictions vs actuals for December 2025 (window 8)



MCS Overall

Model	MCS overall	MCS benchmarks	$F_{\beta=2}$ overall
LightGBM	12.50%	**62.50%**	87.50%
ARIMA	0.00%	37.50%	12.50%
Ridge regression	0.00%	0.00%	37.50%
Naive	0.00%	0.00%	0.00%
Ridge seasonal	0.00%	0.00%	**100.00%**
LSTM	0.00%	50.00%	25.00%
NBEATSx	**75.00%**	–	50.00%
NHiTS	50.00%	–	75.00%
TFT	37.50%	–	62.50%

Table 4: First column shows MCS inclusion rates for all models computed based on TSO and direction datasets ($\alpha = 0.25$). The middle column reports MCS inclusion rates for benchmarks only. The right column reports the percentage of datasets in which each model is dominant according to the $F_{\beta=2}$ pairwise bootstrap superiority test ($\varepsilon = 0.05$). Best values are shown in bold.



Overall conclusions

- Deep neural models appear most frequently in the MCS, with NBEATSx in 6/8 datasets.
- LightGBM is selected most of the time as part of benchmark MCS, with LSTM second place. It is also the only benchmark part of the overall MCS
- Event detection is dominated by the seasonal ridge benchmark, suggesting that occurrence dynamics are partly captured by simple seasonal structure. LightGBM, NHiTS and TFT follow
- For future diagnostics, we choose only LightGBM and LSTM from the benchmark category



Per horizon MCS

- Predictions have been split into horizons $h \in [1, H]$ and MCS has been computed separately

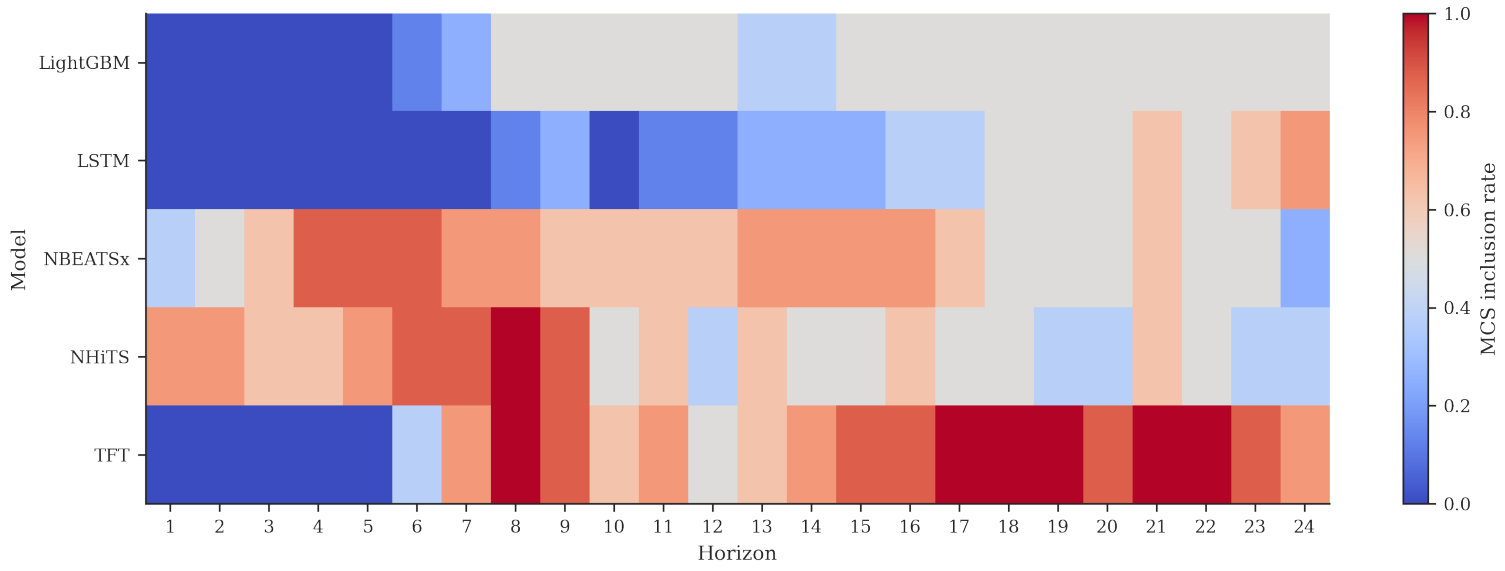


Figure 7: Heatmap of MCS inclusion rates computed per horizon for each model



Per horizon $F_{\beta=2}$

- Predictions have been split into horizons $h \in [1, H]$ and block bootstrap test dominance probabilities have been computed separately

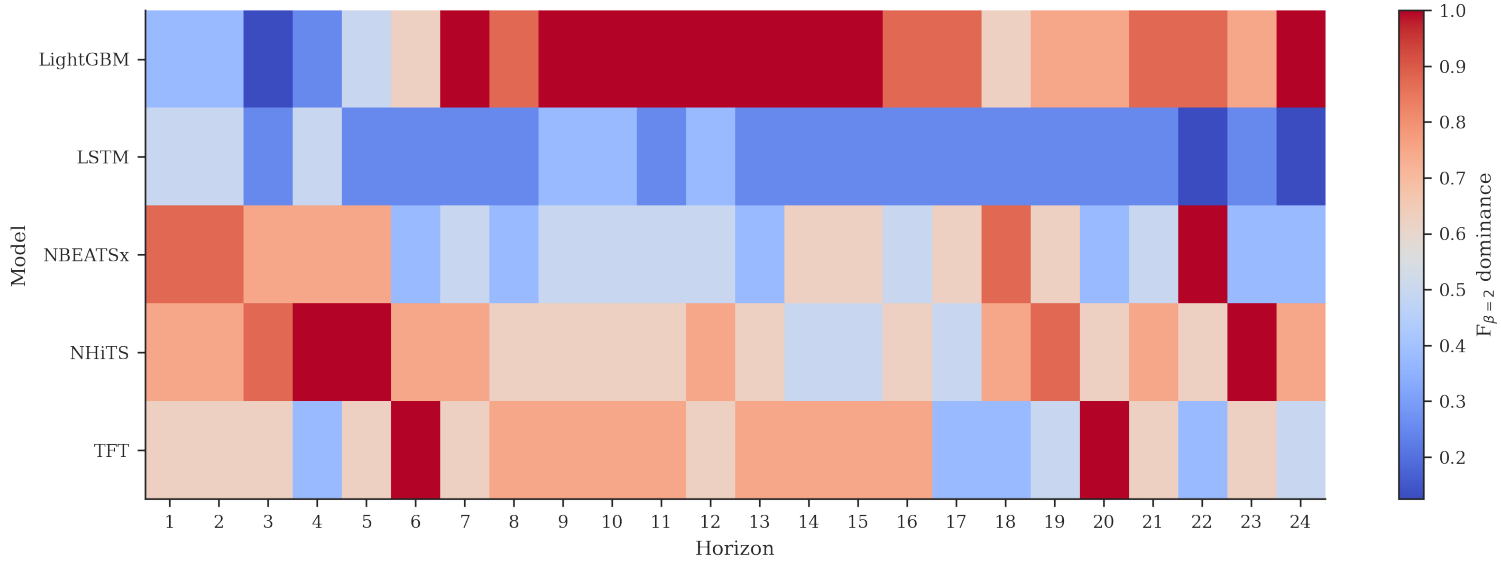


Figure 8: Heatmap of $F_{\beta=2}$ inclusion rates computed per horizon for each model (block bootstrap of length 24, 1,000 blocks)



Per horizon conclusions

- In the early horizon hours, NBEATSx, NHiTS, and TFT are included in the MCS more often (particularly the first), thus they form the best model set.
- For later hours, only TFT is included, which could imply that it is the only best model in this category
- For recall detection, some models shine: neural models early in the morning and late in the evening, while LightGBM most of the day



Regime description

Label	Period	Description
High PV	March 2025	Exceptionally sunny (Bundesnetzagentur, 2025a) with more PV curtailment
PV DSO	Q2 2025	69% more renewable redispatch was reported due to the distribution system YoY (Bundesnetzagentur, 2025b)
PV & Wind	Q3 2025	Renewable curtailment surge by 29% YoY with mixed solar and wind drivers (Bundesnetzagentur, 2026)
Winter	01.10.2025 - 20.06.2026	Harsher and colder winter of 2026 compared to 2024/2025 (Wetterdienst, 2026)

Table 5: Test regime split description



Per regime evaluation

Model details		Regimes			
Model	Metric	High PV	PV DSO	PV & Wind	Winter
LightGBM	MCS	50.00%	37.50%	25.00%	25.00%
	$F_{\beta=2}$	25.00%	62.50%	87.50%	**87.50%**
LSTM	MCS	25.00%	0.00%	0.00%	0.00%
	$F_{\beta=2}$	62.50%	0.00%	50.00%	37.50%
NBEATSx	MCS	50.00%	**62.50%**	62.50%	62.50%
	$F_{\beta=2}$	87.50%	**100.00%**	25.00%	50.00%
NHiTS	MCS	62.50%	37.50%	62.50%	**87.50%**
	$F_{\beta=2}$	**100.00%**	50.00%	**100.00%**	62.50%
TFT	MCS	**75.00%**	50.00%	**87.50%**	75.00%
	$F_{\beta=2}$	50.00%	25.00%	62.50%	75.00%

Table 6: Per-regime performance summary. MCS inclusion rates ($\alpha = 0.25$) and $F_{\beta=2}$ dominance rates are reported for each model. Best values per regime and metric are shown in bold.



Per regime conclusions

- No forecaster is able to dominate even for one metric in each regime
- For MAE, the MCS top contender is traded between NBEATSx (April - June 2025), TFT (March 2025, Q3 2025), and NHiTS (winter regime)



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Methodology

- *Do exogenous covariates significantly improve performance?*
  $\Downarrow$
- We train no-covariate models and test for significant differences with DM.
- *Is their predictive power driven by correct temporal alignment?*
  $\Downarrow$
- We construct a placebo by circularly shifting all covariates: $\tilde{x}_t = x_{(t+K) \bmod T}$, where $K = 17 \cdot 24$ hours, which preserves hourly structure but breaks weekly alignment with the target
- This approach keeps most covariate properties intact, but breaks alignment with the target



Placebo

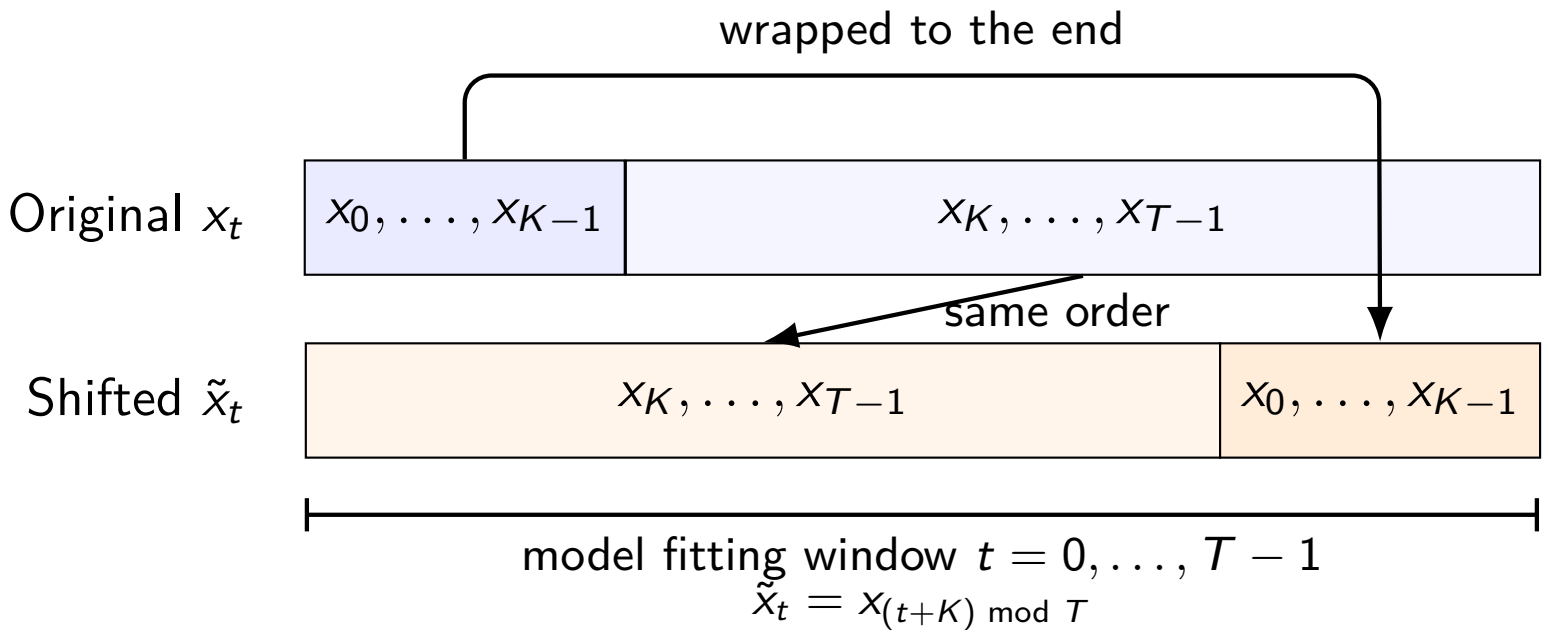


Figure 9: Circular shift used in the robustness check. The covariate sequence is shifted forward by K time steps while preserving the same fitting window length T.



Methodology

- Sparse directions are reported separately because the approximation $\hat{y} \approx 0$ is competitive and does not require covariates to be learned
- For computational tractability, only NHiTS is chosen as a representative model
- The model is trained on 2 separate seeds



Robustness results

Comparison details		Seed 1		Seed 2	
Comparison	Type	Unexp	Exp	Unexp	Exp
No Cov vs Full Cov	Dense	0.00%	60.00%	0.00%	60.00%
	Sparse	33.33%	33.33%	0.00%	0.00%
No Cov vs Shift	Dense	20.00%	0.00%	40.00%	0.00%
	Sparse	0.00%	0.00%	0.00%	0.00%
Shift vs Full Cov	Dense	0.00%	60.00%	0.00%	40.00%
	Sparse	0.00%	0.00%	0.00%	0.00%

Table 7: Diebold-Mariano (DM) dominance frequencies across TSOs and directions. “Exp” denotes rejections in the expected direction under the robustness hypothesis: Full Cov outperforms No Cov and Shift. For No Cov vs Shift, expected differences are not directional and are reported mainly as a placebo comparison. "Unexp" denotes rejections in the opposite direction. Results are reported separately for dense and sparse regimes and for two random seeds. Sparse regimes correspond to series with more than 70% zero observations.



Robustness conclusions

- In the majority of dense cases (40–60%), the full-covariate specification significantly outperforms both the no-covariate and shifted variants
- This confirms the presence of genuine predictive signal in the covariates (Hypothesis 1) and demonstrates that their temporal alignment is crucial (Hypothesis 2)
- The shifted and no-covariate models exhibit broadly comparable performance, but there are hints at the power of features, even if signal is not correctly aligned



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Setup

- Integrated Gradients (IG) is used (Sundararajan, Taly, & Yan, 2017) with standardized mean as baseline
- Absolute (unsigned) attributions are computed; historical and future contributions are summed
- Aggregation per feature type is conducted
- A priori, we expect lags, production/consumption, wind/PV, scheduled exchanges to be important
- TenneT DE and 50Hertz specifically should be less dependent on lags



Overall

Dataset details		Integrated-gradient feature groups		
TSO	Direction	Dominant group	Second group	Third group
50Hertz	down	Lags (61%)	Prod./cons. (18%)	SCE (8%)
50Hertz	up	Lags (42%)	Prod./cons. (27%)	SCE (11%)
Amprion	down	Lags (34%)	Prod./cons. (29%)	SCE (15%)
Amprion	up	Lags (47%)	Prod./cons. (24%)	SCE (10%)
TenneT DE	down	Prod./cons. (32%)	Lags (32%)	SCE (13%)
TenneT DE	up	Lags (34%)	Prod./cons. (31%)	SCE (12%)
TransnetBW	down	Prod./cons. (30%)	Lags (27%)	SCE (17%)
TransnetBW	up	Prod./cons. (31%)	Lags (25%)	SCE (16%)

Table 8: Top three dominant feature groups according to integrated-gradient attributions for each TSO and redispatch direction. "Prod./cons." denotes production/consumption features; and "SCE" denotes scheduled commercial exchanges.



Per Horizon

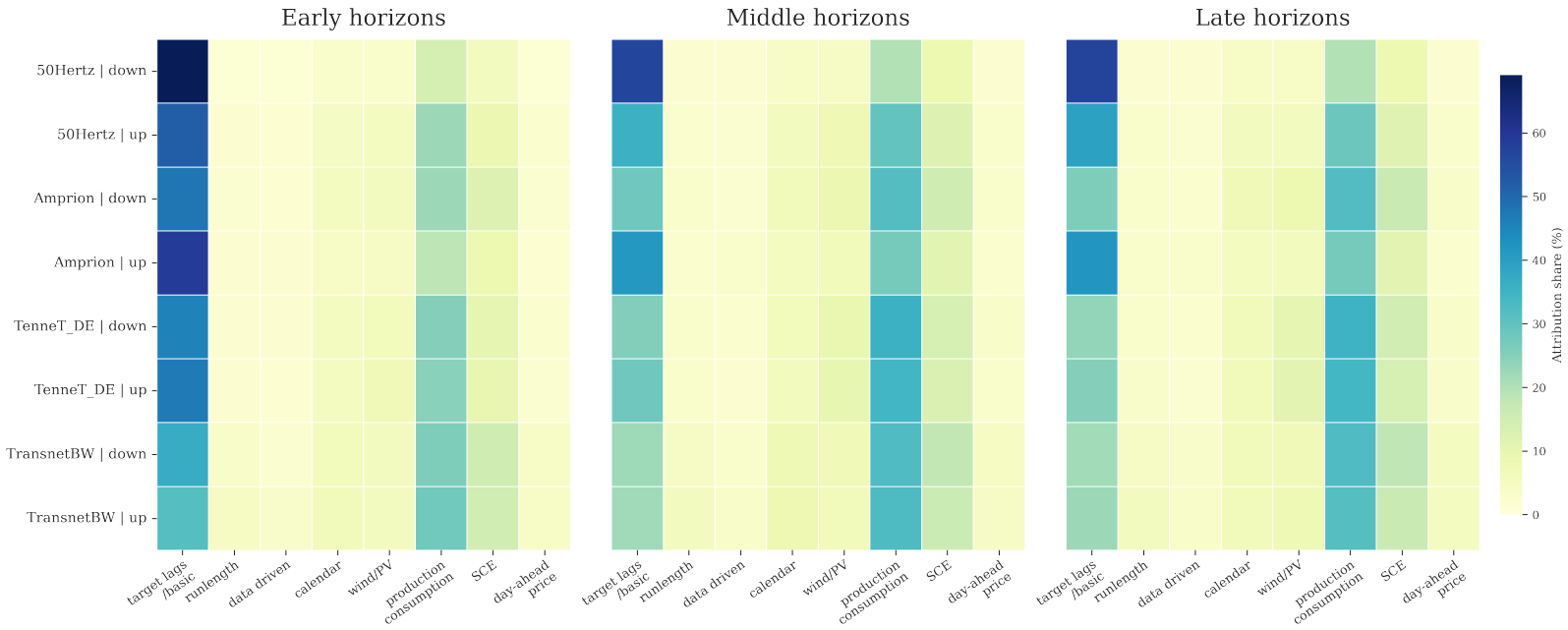


Figure 10: NHiTS IG attributions of early (1–9 hours), mid (10–17 hours), and late (18–24 hours) horizon per feature type, for each TSO, on the full test set.



Conclusions

- Past values are deemed crucial in most cases, even 50Hertz, which is a bit unexpected but can be explained by its cooperation to other TSOs
- Sparse datasets show weaker and less stable attribution patterns, possibly reflecting joint direction training and the dominance of zero periods.
- TenneT DE relies more on prod/cons, particularly at longer horizon hours



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Takeaways

- A methodology for ML forecasters for redispatch has been presented
- Overall, results indicate that deep neural forecasters dominate, each excelling in a different regime
- Signal decomposition forecasters tend to more accurately predict the first hours of the horizon, while TFT wins on the latter ones
- Relevant exogenous variables/covariates have been shown to enhance performance for NHiTS, which may generalize to NBEATSx and TFT, but should be verified directly



Operational impact

- We expect our proposed approach to complement existing operational practices given specialized data
- Next steps:
  - Assessing economic cost in a controlled environment, such as a bus system (similar to optimization studies)
  - Probabilistic forecasting for proper risk management (e.g., 90th percentile)
  - More complex forecasters (LLM's, foundational models) or architectures tailored to this task



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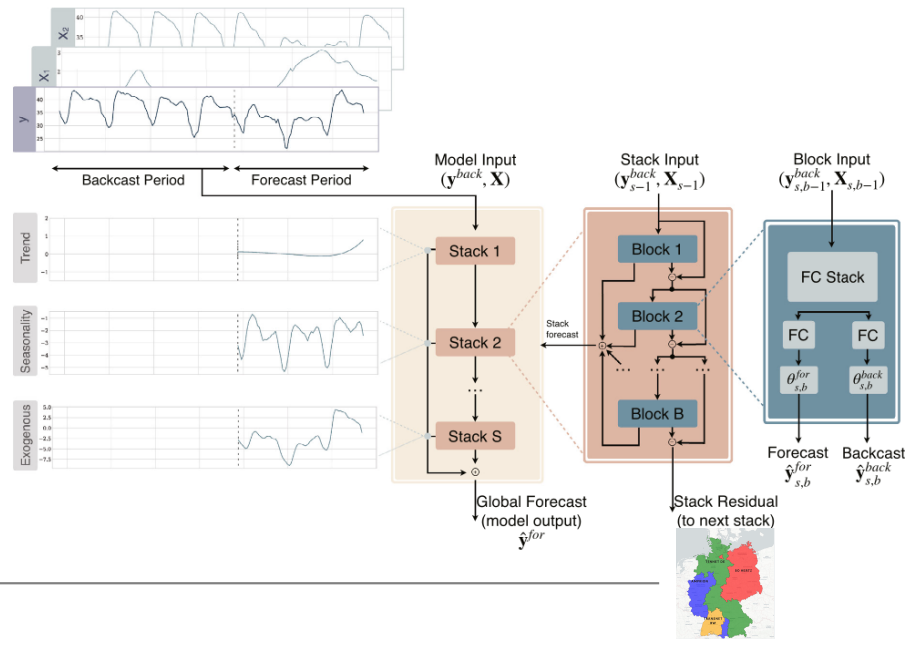
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N-BEATS Description

- Introduced in Oreshkin et al., 2020
- Black-box or interpretable architecture
- Supports transfer learning
- Decomposes signal into *backcast* (history) and *forecast* (prediction)

- Two main building blocks:
  - *Blocks*: capture specific information using Fully-Connected (FC) layers and a learnable linear projection
  - *Stacks*: a collection of blocks



Back to "Model selection"

NBEATSx Model Architecture

$$
\begin{aligned}
h_{s,b} &= \text{MLP}_{s,b}\left(y_{\text{back},s,b-1}, X_{s,b-1}\right) \\
\theta_{\text{back},s,b} &= \text{Linear}_{\text{back}}\left(h_{s,b}\right) \\
\theta_{\text{for},s,b} &= \text{Linear}_{\text{for}}\left(h_{s,b}\right)
\end{aligned}
$$

Basis Expansion:

$$
\begin{aligned}
\hat{y}_{\text{back},s,b} &= V_{\text{back},s,b}\theta_{\text{back},s,b} \\
\hat{y}_{\text{for},s,b} &= V_{\text{for},s,b}\theta_{\text{for},s,b}
\end{aligned}
$$



NBEATSx Model Architecture

Stack and Residual Update:

$$y_{\text{back},s,b+1} = y_{\text{back},s,b} - \hat{y}_{\text{back},s,b}$$

$$\hat{y}_{\text{for},s} = \sum_{b=1}^{B} \hat{y}_{\text{for},s,b}$$

where:

- $y_{\text{back},s,b}$ is the backcasted time series input.
- $X_{s,b}$ represents the exogenous variables used in the block.
- $h_{s,b}$ is the hidden representation in block b of stack s.
- $\theta_{\text{back},s,b}$ and $\theta_{\text{for},s,b}$ are the expansion coefficients.
- $V_{\text{back},s,b}$ and $V_{\text{for},s,b}$ are the learned basis functions.



NBEATSx

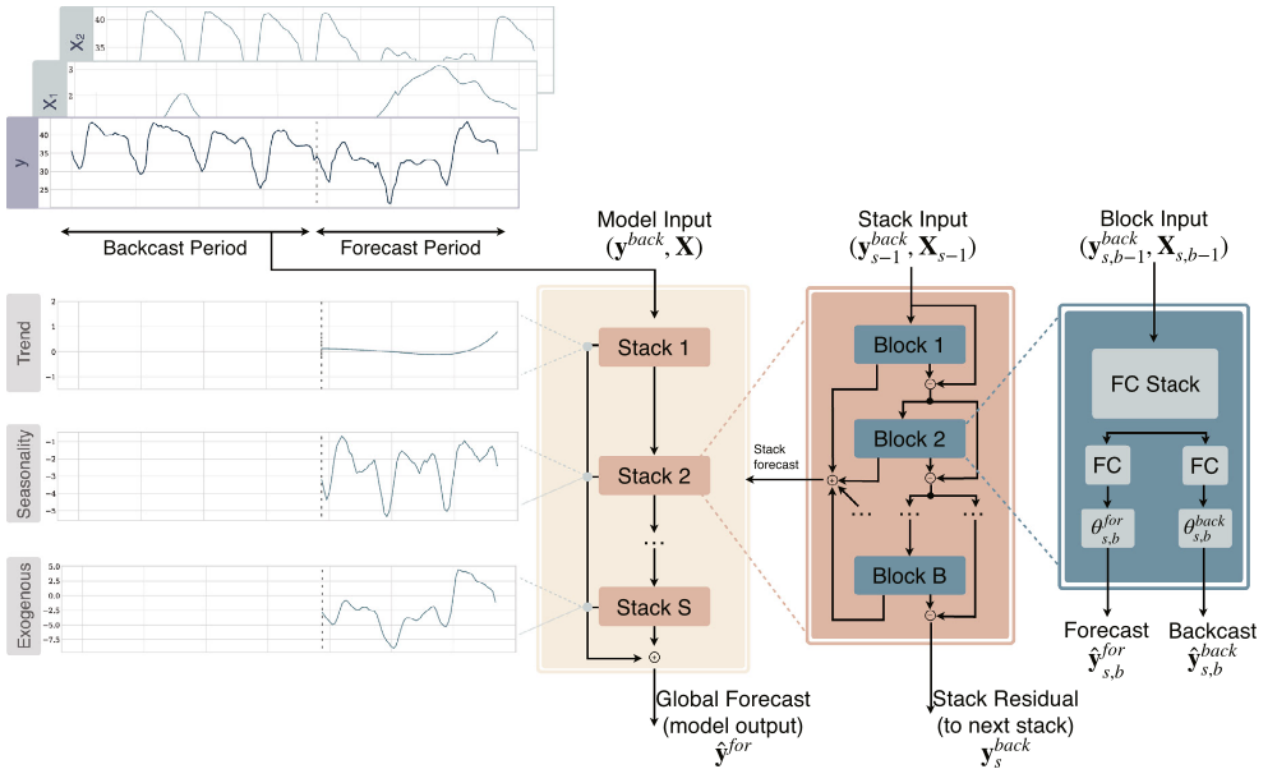


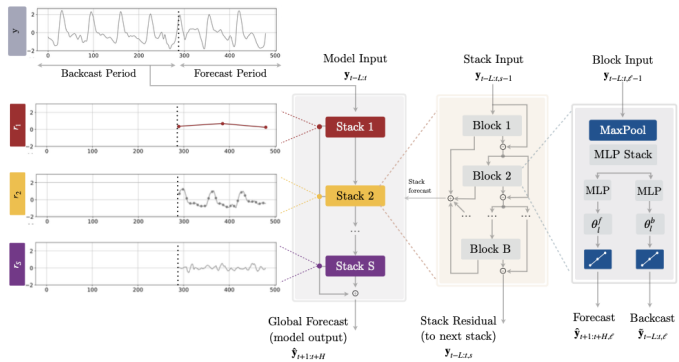
Figure 11: NBEATS with exogenous variables, Olivares et al. 2024



N-HiTS Description

- Introduced in Challu et al., 2022
- An evolution of N-BEATS by:
  - Included kernel pooling at each block's entry point
  - Regularized basis function forms by hierarchical interpolation and multi-rate data sampling

- Unique parameters from N-BEATS:
  - *Pooling*: control how pooling is done (max, mean, etc.), and how much the inputs are shrinked
  - *Interpolation type*: three were proposed in the paper (linear, cubic, and nearest neighbor)





NHITS Architecture

NHITS model consists of multiple blocks using hierarchical interpolation and multi-rate sampling:

$$h_\ell = \text{MLP}_\ell\left(\text{MaxPool}\left(y_{t-L:t,\ell}, k_\ell\right)\right)$$
$$\theta_f^\ell = \text{Linear}_f(h_\ell)$$
$$\theta_b^\ell = \text{Linear}_b(h_\ell)$$

Hierarchical Interpolation:

$$\hat{y}_{\tau,\ell} = g(\tau, \theta_f^\ell), \quad \forall \tau \in \{t+1, ..., t+H\}$$
$$\hat{y}_{\tau,\ell} = g(\tau, \theta_b^\ell), \quad \forall \tau \in \{t-L, ..., t\}$$



NHITS Architecture

Final Forecast Assembly:

$$\hat{y}_{t+1:t+H} = \sum_{\ell=1}^{L} \hat{y}_{t+1:t+H,\ell}$$

$$y_{t-L:t,\ell+1} = y_{t-L:t,\ell} - \tilde{y}_{t-L:t,\ell}$$

where $y_{t-L:t,\ell}$ is the input time series segment with lagged observations, k_ℓ is the kernel size for multi-rate sampling, h_ℓ is the hidden representation in block ℓ. θ_f^ℓ and θ_b^ℓ are the forward and backward interpolation coefficients, $g(\tau, \theta_f^\ell)$ is the hierarchical interpolation function mapping latent features to future predictions, $\hat{y}_{t+1:t+H}$ is the final assembled forecast by summing the block-wise outputs.



NHITS Model Illustration

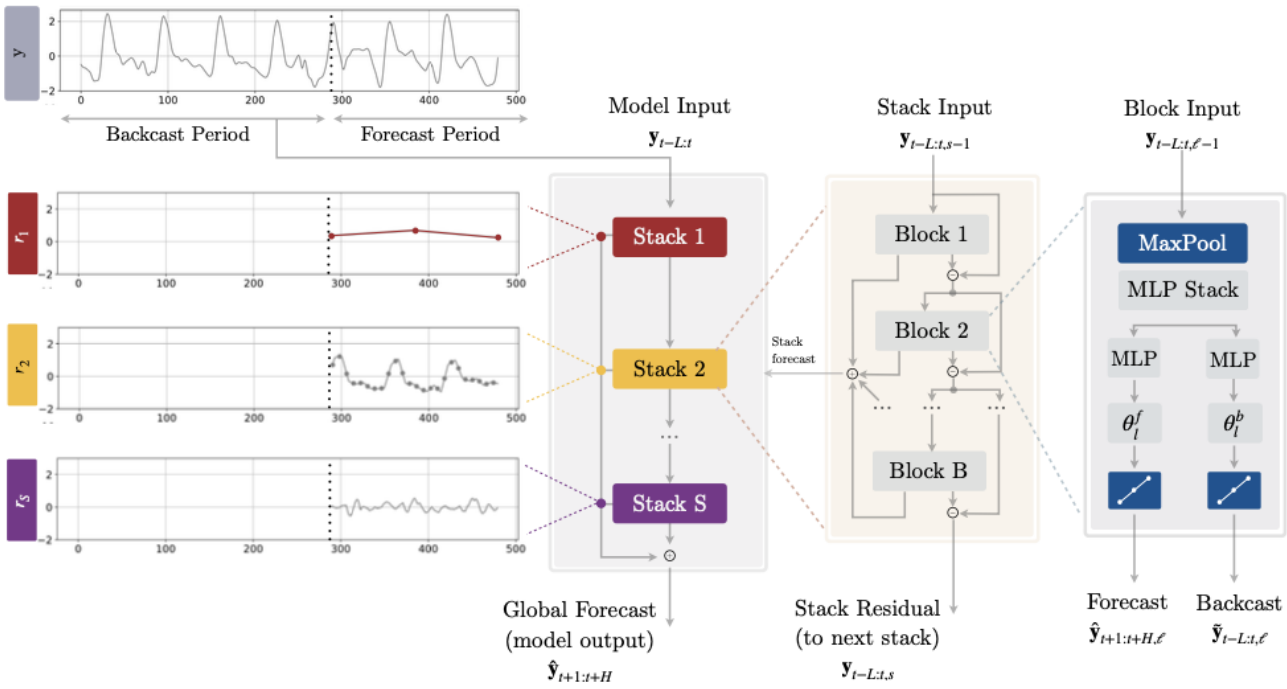


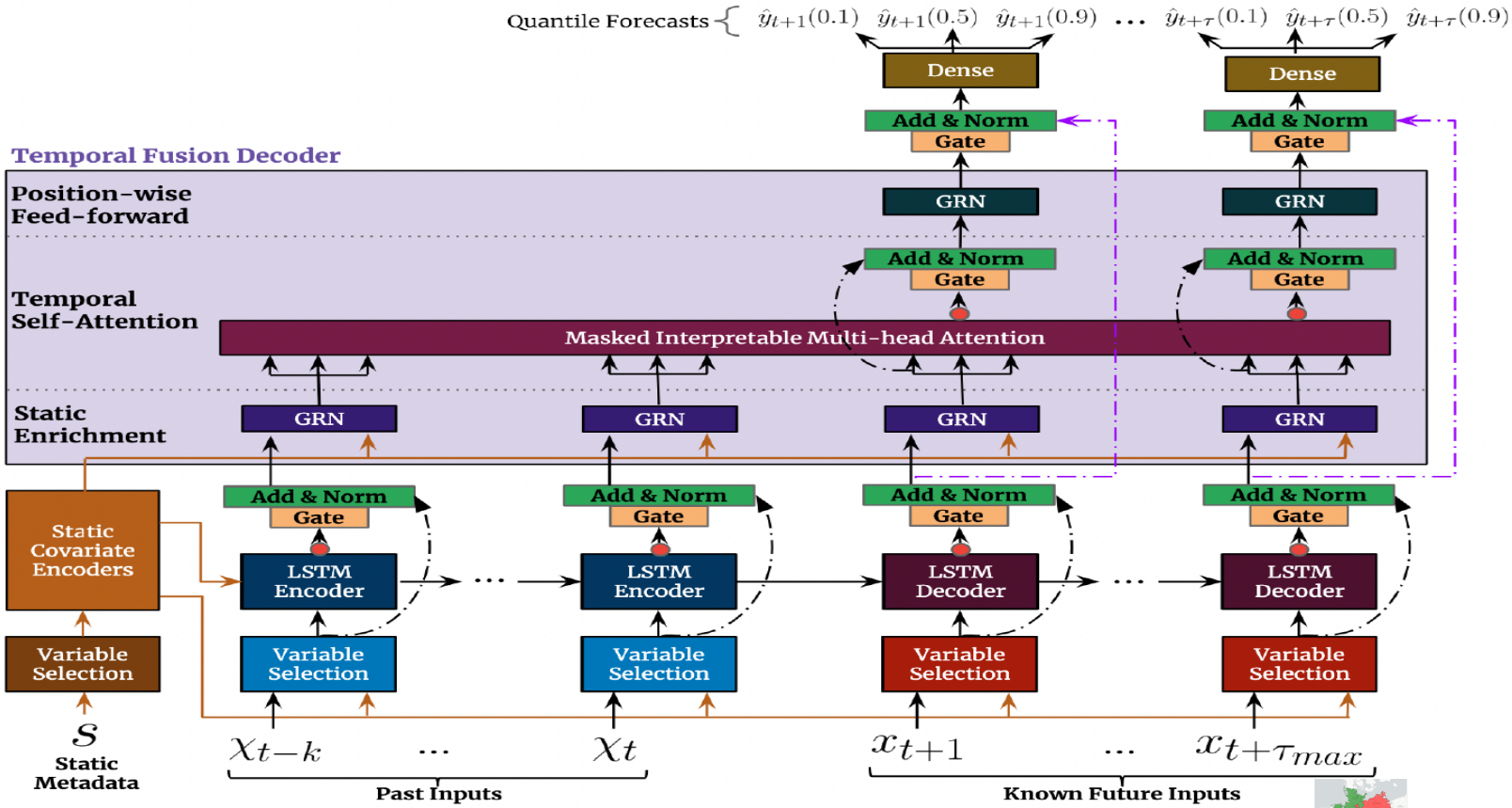
Figure 12: NHITS model with multi-rate input sampling and hierarchical interpolation.

▸ Back to "Model selection"



TFT Description

- Introduced in Lim et al., 2021



TFT Model Architecture

Mathematical Formulation:

$$
\begin{aligned}
h_t &= \text{LSTM}\left(y_{t-L:t}, X_{t-L:t}\right) \\
\theta_t^{\text{enc}} &= \text{VariableSelection}(X_t) \\
\theta_t^{\text{dec}} &= \text{MultiHeadAttention}(Q, K, V)
\end{aligned}
$$

Self-Attention Mechanism:

$$
A(Q, K) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)
$$

$$
H = \sum_{h=1}^{H} A(QW_Q^{(h)}, KW_K^{(h)})VW_V^{(h)}
$$



TFT Model Architecture

Final Forecasting Step:

$$\hat{y}_{t+1:t+H} = \sum_{b=1}^{B} \hat{y}_{t+1:t+H,b}$$

where:

- $y_{t-L:t}$ represents the past observations.
- $X_{t-L:t}$ contains past and future covariates.
- h_t is the hidden state representation.
- θ_t^{enc} and θ_t^{dec} are feature selection weights.
- $A(Q, K)$ computes the attention scores.
- H is the multi-head attention output.
- $\hat{y}_{t+1:t+H}$ is the final forecast.

▸ Back to "Model selection"



Market-based congestion relief

Service	Description	Activation	Implementation
Market coupling	Cross-zonal optimization via prices	Implicit	EUPHEMIA
LMP / DLMP	Locational pricing reflecting congestion	Implicit	Market design
Market-based redispatch	Congestion relief via bids	Semi-auto	TSO market platforms
P2P trading	Local energy exchange	Implicit	Decentralized markets

Table 9: Different types of congestion management mechanisms at the market level. Inspiration: (Zhou, Tesfatsion, & Liu, 2011), (Staudt, Rausch, et al., 2019), (Doan et al., 2025)

▸ Back to "Motivation"



Physical congestion relief

Service	Description	Activation	Implementation
FACTS devices	Power flow control devices	Automatic	Grid-integrated
DLR	Dynamic adjustment of line capacity	Automati	Weather-based systems
Grid expansion	Increase transmission capacity	Long-term	Infrastructure projects
Transformer control	Tap changers / phase shifting	Automatic	Substation equipment

Table 10: Different types of congestion management mechanisms at the physical level. Inspiration: (Yousefi et al., 2012), (Mansour Saatloo et al., 2022), (Emery, Karpatchev, & Tchoubraev, 2005)

▸ Back to "Motivation"



Flexibility congestion relief

Service	Description	Activation	Implementation
Demand response	Adjust consumption patterns	Semi-auto	Aggregators
Energy storage (ESS)	Shift energy across time	Automatic	Batteries / storage systems
Sector coupling	Cross-sector energy conversion (e.g. H_2)	Automatic	Integrated systems
Flexible generation	Dispatchable generation response	Automatic	Hydro / gas plants

Table 11: Different types of congestion management mechanisms on the flexibility side. Inspiration: (Shabbir et al., 2024), (Bernecker et al., 2025), (Khajeh, Seyyedeh-Barhagh, & Laaksonen, 2025)

▸ Back to "Motivation"



Operator-based congestion relief

Service	Description	Activation	Implementation
Cost-based redispatch	TSO-controlled generation adjustment	Manual	National frameworks
Topology switching	Network reconfiguration	Manual	Control room operation
Preventive scheduling	Forecast-based congestion avoidance	Manual	TSO planning tools

Table 12: Different types of congestion management mechanisms at the operator (TSO/DSO) level. Inspiration: (Poplavskaya et al., 2020), (Kloubert et al., 2015), (Liao et al., 2025)

▸ Back to "Motivation"



Training Information Set

- Consider S the gap between avaialable and real timestamps. What we want is to predict $y_{t+1:t+H}$ using only information available at decision time $t - S$
- During training target and future covariates are shifted forward to define a model-time problem:

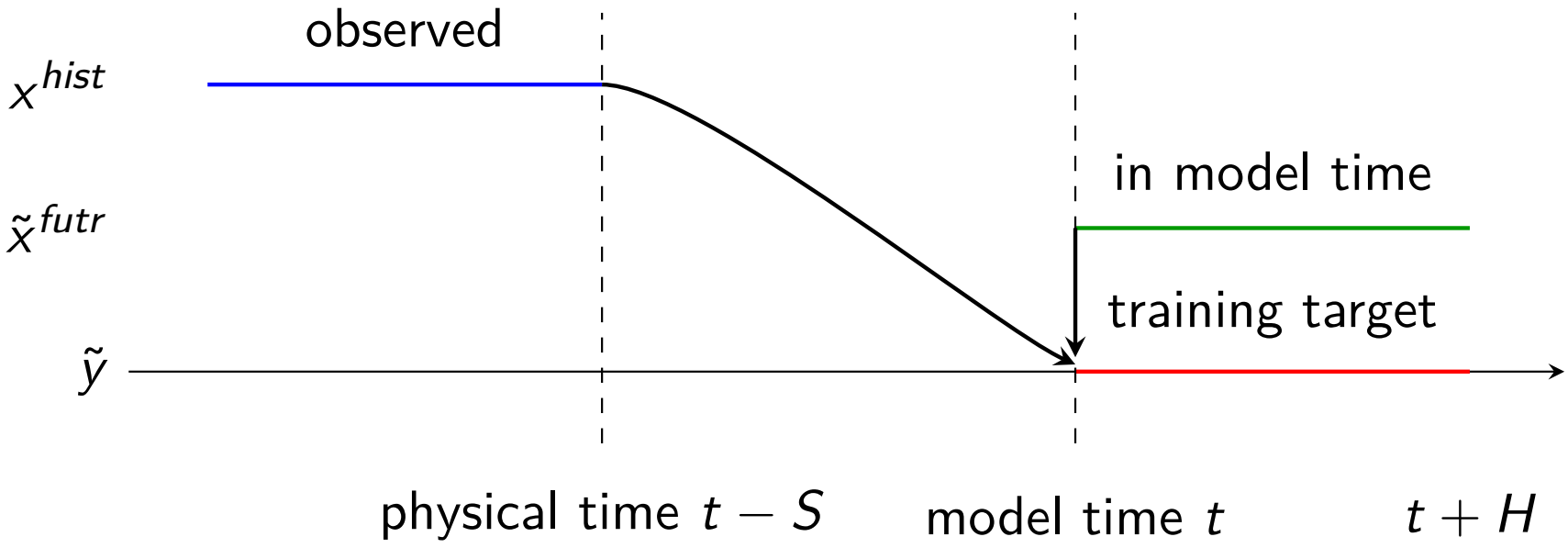
$$\tilde{y}_t = y_{t+S}, \qquad \tilde{x}_t^{futr} = x_{t+S}^{futr}$$

- Historical covariates x^{hist} are not shifted, since they must reflect only information available up to the decision time
- The model is therefore trained on shifted targets, but evaluated and used in physical time

▸ Back to "Forecaster setup"



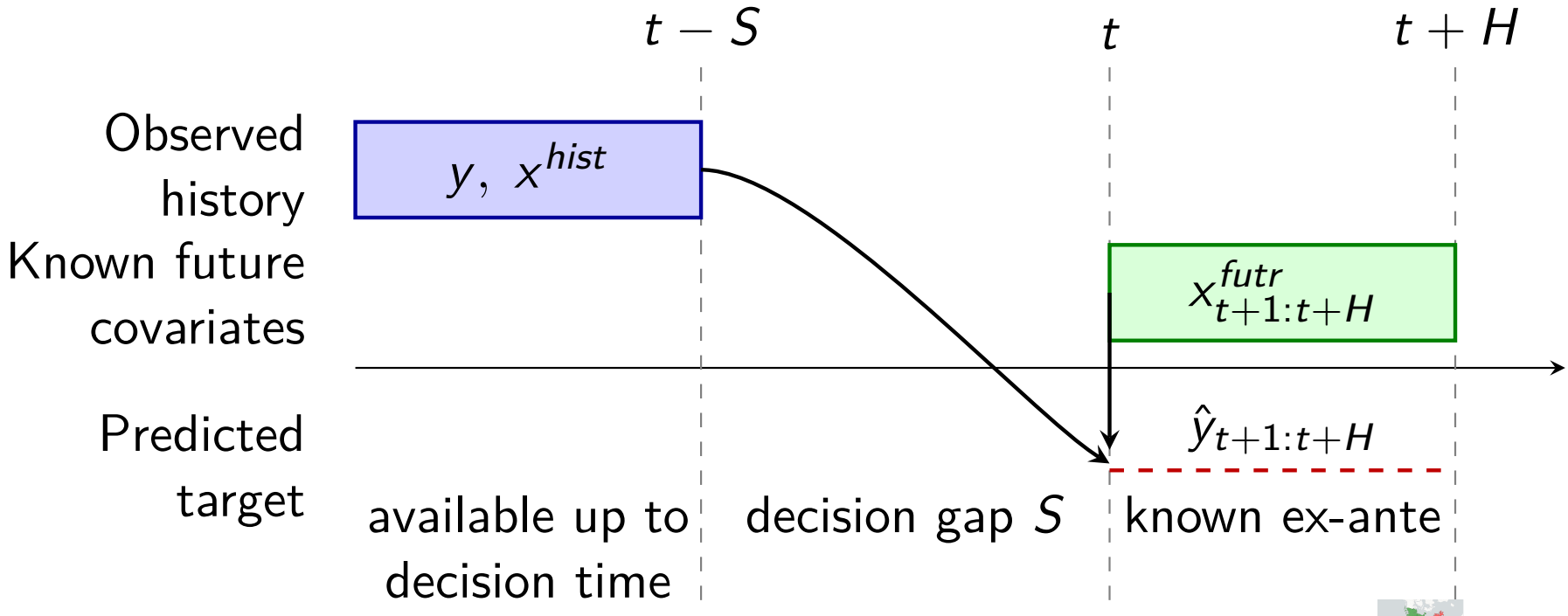
Training information set: data alignment





Inference: available information and forecast horizon

- At inference, inputs are aligned consistently, forecasts are generated in model time, and then mapped back to physical time



Hourly aggregation

- Each intervention i, we are given a timeframe (t_i, T_i) and the average energy in MWh $E_i = 1/4 \sum_t l_{i,t}$ associated with unknown 15min increments $l_{i,t}, t \in [t_i, T_i]$.
- For each hour h, we compute the overlap of that hour with the intervention span:

$$overlap_{i,h} = min(T_i, h+1) - max(t_i, h)$$

- And then convert into continuous hourly signal:

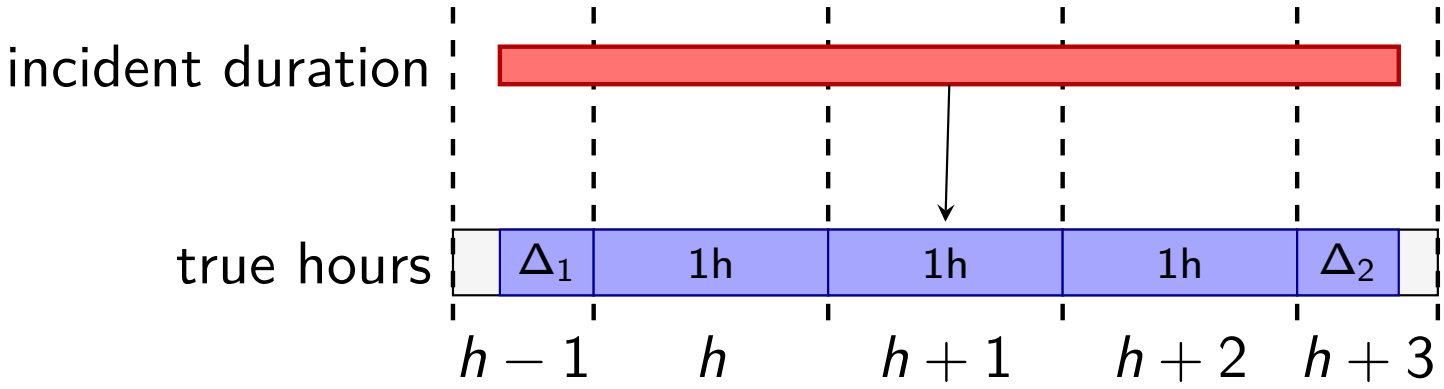
$$E^{eq}_{i,h} = E_i \cdot \frac{overlap_{i,h}}{T_i - t_i}$$

- The approach above assumes constant output for the entire incident period, which is not always the case.

▸ Back to "Target construction"



Hourly overlap construction





Hyperparameter tuning

- 3 development windows are used, with testing period [2025-01-01, 2025-03-01)
- For each model, one or two influential parameters are used for training in a coarse grid, and combinations within $\tau = 2\%$ of the best performing model on both volume-weighted MAE and $F_{\beta=2}$ are kept
- A parsimony rule is applied to survivors for selecting the best combination
- The process is repeated for each TSO
- Input size is set globally by majority voting for all models using the same procedure

▸ Back to "Training Overview"

